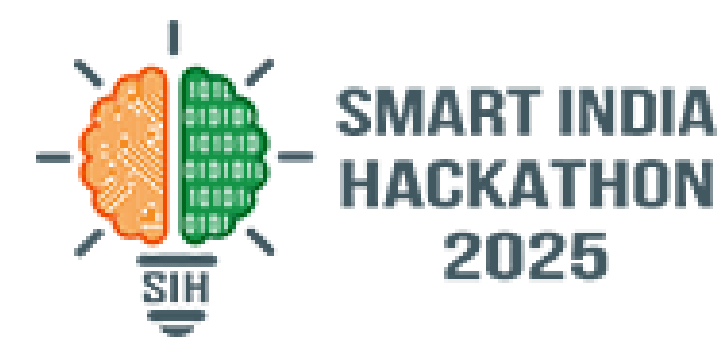


SMART INDIA HACKATHON 2025



- **Problem Statement ID** – SIH25002
- **Problem Statement Title** – Student Innovation:
Swadeshi for Atmanirbhar Bharat - Blockchain
& Cybersecurity
- **Theme** - Blockchain & Cybersecurity
- **PS Category**- Hardware
- **Team ID**- 77332
- **Team Name**- Team Ataltra





Idea/ Solution

Proposed Solution: ExploreSure is a Swadeshi Hardware & DLT Security System. It uses a Compact IoT Identity Tag to generate a tamper-proof Digital ID, empowering agencies with proactive, hardware-level security.



Uniqueness/Innovation

- **Core Innovation (Hardware):** Low-Cost IoT Tag (3x3x1 cm) with LoRa/LoRaWAN for extreme remote connectivity. Guarantees Digital ID assurance (hardware-based security).
- **Atmanirbhar Hybrid Security:** Permissioned DLT verifies immutable IDs. AES-256 encryption secures data transmitted from the hardware. Robust, viable, and sovereign security model.
- **Proactive Sentinel:** AI uses data from the Tag to detect threats, making the hardware predictive, not just reactive.



Complete journey

ExploreSure - Swadeshi Digital Sentinel





Architecture

Centered on the IoT Smart Tag and its integration into a DLT network. Built for robustness and economic viability.



Components

- Processor (MCU): ESP32 Pico D4 (Runs firmware, encrypts data).
- Location: NEO-8M GPS/GNSS (Captures precise coordinates).
- Comms: LoRa/LoRaWAN Module (Ensures data transmission in remote, low-connectivity zones).
- Form Factor: Custom 3D-Printed Housing (Compact size: 3x3x1 cm).

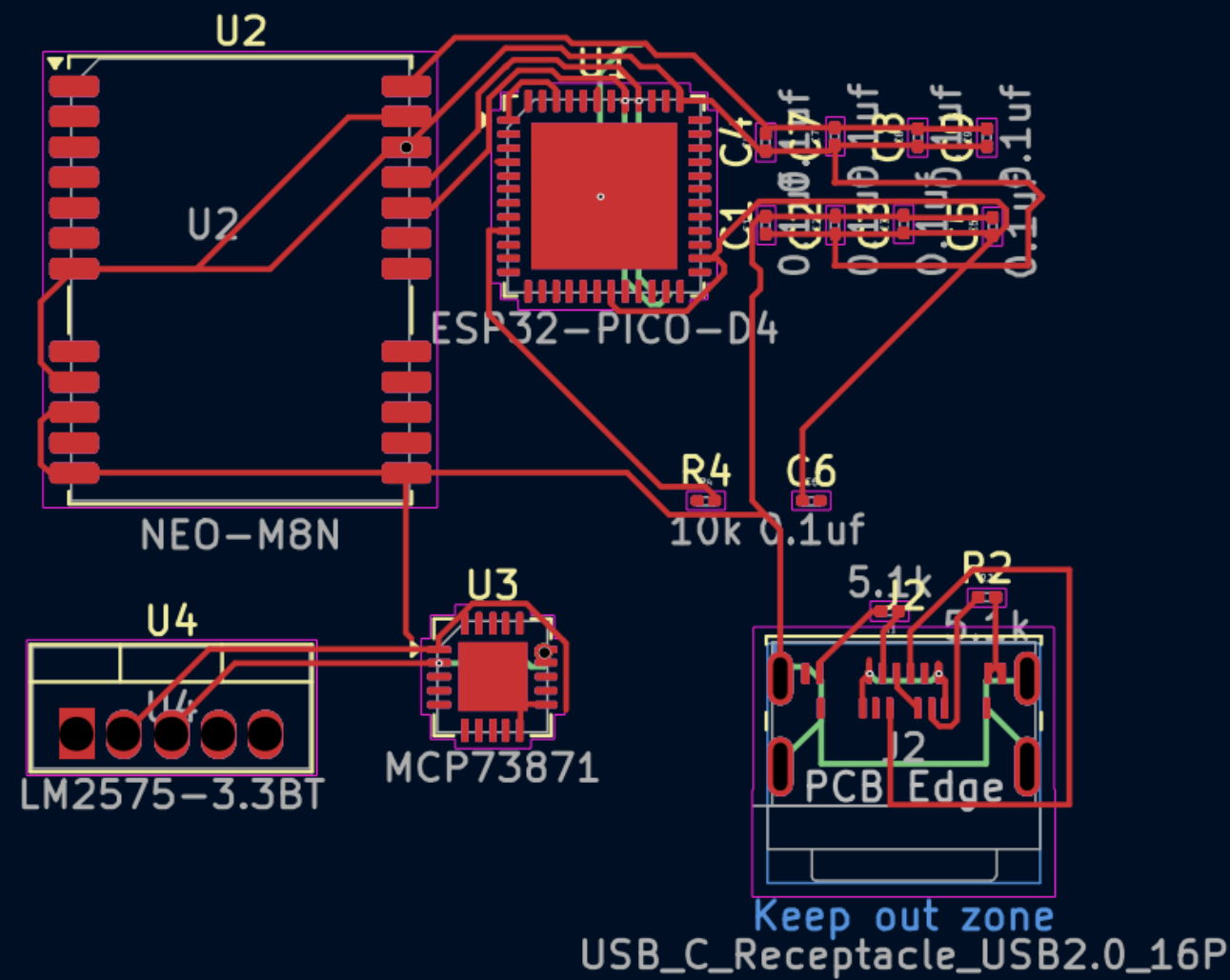


Methodology

Capture > Analyse > Alert > Respond.

- **Hardware & Comms:** IoT Tag (Hardware): ESP32, LoRa/LoRaWAN, Low-Power GPS.
- **Identity & Security:** DLT Security: Permissioned Blockchain & AES-256 Encryption.
- **Proactive Intelligence:** AI/ML: Python and Scikit-learn power anomaly detection.
- **Backend & Logic:** Full Stack Core: Node.js, Express.js, Firebase/MongoDB manage data/API.
- **Interface & Viz:** Frontend/UI: React, HTML, and Tailwind CSS build dashboards.

PCB Artchitecture of EXPLORESURE !!





Feasibility Analysis

- **Feasibility:** A working Hardware/Software Prototype is feasible, centered on the operational IoT Tag.
- **Integration:** Designed for low-cost, domestic manufacturing to achieve Atmanirbhar goals rapidly.
- **Scalability:** DLT model and microservices handle national deployment and high-volume data streams.



Potential Challenges

- **DLT Security/Compliance** (Immutability vs. Data Privacy).
- **Remote Data Integrity** (Low-connectivity zones).
- **Hardware Battery Life** (Optimizing operation remotely).
- **Agency Deployment Simplicity** (Ease of government use).



Overcoming Challenges

- **LoRa/LoRaWAN** for robust remote data assurance.
- **National Asset Tracking** for long-term project viability.
- Simple UI/Deployment for government personnel.
- **DIV API Licensing** for B2G revenue stream.



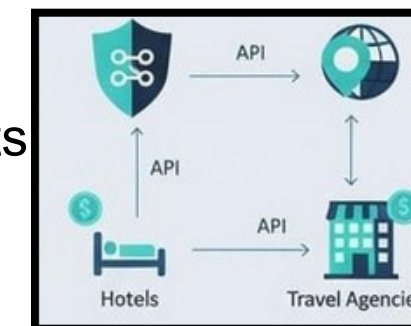
Revenue model



Government Subscriptions
– Annual B2G fees for Proactive Digital Sentinel services.



Digital Insights Licensing:
Selling anonymized data insights for security/infrastructure planning.



DIV API Licensing: Licensing the Digital Identity Verification (DIV) API and DLT protocols to third-party government bodies.



Potential Impacts

- **Digital Sovereignty & Security:** Provides a secure, indigenous, and auditable identity framework for critical national zones.
- **Border & Critical Zone Agencies:** Protecting sensitive areas with proactive, hardware-verified monitoring (DLT Tag data).
- **Local Economic Empowerment:** Boosts local manufacturing of IoT components and increases visitor confidence in remote areas.



Benefits

- **Proactive Digital Defense:** AI flags behavioral threats from the Tag, giving agencies a crucial lead time in security and emergency response.
- **Strengthened Governance:** Authorities gain smart, verifiable DLT tools to improve public safety and enforce critical zone trespassing.
- **Atmanbharat & Economic Impact:** Builds foundational indigenous DLT/AI technology, boosting domestic manufacturing and reducing foreign surveillance reliance.

References

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- **Evaluation of Blockchain Technology in the Context of Tourism Sector...Challenges and Opportunities**

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